

Supporting Information

GVIM 2.0: A Multi-Agent Research System for Chemistry and Materials Science

Kangyong Ma

College of Physics and Electronic Information Engineering, Zhejiang Normal University, Jinhua 321000, China

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S1. Evaluation datasets and endpoints

This Supporting Information summarizes evaluations based on archived GVIM artifacts and public benchmark source data. The evaluation set includes public question-answer benchmarks, task-native workflows with externally defined gold standards, and retrospective discovery or temporal-validation studies based on public labels. Reported endpoints include answer-key accuracy, prediction error, extraction F1, retrieval accuracy, recall, enrichment factor and external-validation metrics.

Table S1. Evaluation datasets and reported endpoints.

Evaluation class	Reported endpoint	Evaluation role
Public benchmark	Native answer-key accuracy	Domain question-answer performance under benchmark rules
Task-native workflow	MAE, RMSE, R ² , precision, recall, F1, top-k, MRR	Execution of auditable data-to-result workflows
Retrospective discovery	Recall@k, EF@k, prediction error	Prioritization within a public labelled library
Temporal external validation	External MAE/RMSE/R ² and top-decile recovery	Temporal generalization to later public compounds

S2. Public 400-task benchmark

The benchmark suite contains 250 ChemBench, 97 MaScQA and 53 ChemLLMBench questions. Each source was scored with its native answer key or task-specific rule. Aggregate accuracy across the 400 questions is reported as a descriptive summary, with benchmark-level results presented separately. The same-base comparison used the identical 400 questions for GVIM and the DeepSeek API-only setting.

Table S2. Benchmark composition and same-base results.

Benchmark	n	GVIM	API-only	GVIM (%)	API-only (%)
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Benchmark	n	GVIM	API-only	GVIM (%)	API-only (%)
ChemBench	250	204/250	189/250	81.60	75.60
MaScQA	97	94/97	73/97	96.91	75.26
ChemLLMBench	53	27/53	23/53	50.94	43.40
Aggregate descriptive summary	400	325/400	285/400	81.25	71.25

The paired contingency table contained 268 both-correct, 57 GVIM-only, 17 API-only and 58 both-wrong outcomes. The paired difference was 10.0 percentage points, with a bootstrap 95% CI of 6.0–14.25 percentage points and exact two-sided McNemar $p = 3.40 \times 10^{-6}$.

Table S3. Additional API-only reference runs on the 400-task suite.

Model	Correct	Accuracy (%)	Run status
meta-llama/llama-4-scout-17b-16e-instruct	291/400	72.75	400 completed
llama-3.3-70b-versatile	260/400	65.00	400 completed
qwen/qwen3-32b	245/400	61.25	399 completed; 1 truncated
llama-3.1-8b-instant	169/400	42.25	393 completed; 7 truncated

S3. Task-native chemistry and materials evaluation workflows

Each reported workflow is linked to an archived GVIM front-end thread and was scored against a public or publisher-defined reference. For Matbench steels, the archived front-end evidence corresponds to official fold 0: the front end used internal shuffled five-fold cross-validation on the 249-row fold-0 training partition, refitted the selected model, and predicted 63 unlabeled fold-0 test records. A separate local official-five-fold rerun is documented separately because it followed a different execution route.

Table S4. Archived artifact-supported task-native evaluations.

Workflow	Gold/dataset	Objective result
ESOL solubility prediction	MoleculeNet ESOL	MAE=0.603, RMSE=0.741, $R^2=0.739$; $n_{\text{test}}=254$
Matbench steels fold 0	Matbench v0.1 matbench_steels	MAE=111.444 MPa, RMSE=164.320 MPa, $R^2=0.684$; $n_{\text{test}}=63$
MinerU reaction-table extraction	Publisher JATS XML table nanomaterials-12-01070-t002	Exact precision=0.981, recall=0.981, F1=0.981, row accuracy=0.905
MS/MS candidate retrieval	MassBank InChIKey identity from public MassBank-data records	Top-1=0.800, Top-3=1.000, Top-5=1.000, MRR=0.900; $n_{\text{queries}}=5$
ChEMU entity extraction	ChEMU 2020 Task 1 sample v3	Exact P/R/F1=0.931/0.728/0.817; relaxed F1=0.841; gold spans=92

Workflow	Gold/dataset	Objective result
BACE active discovery	bace_active_discovery.csv	Random Recall@150=0.109, UCB Recall@150=0.363, EF@150=3.664, 95% CI=0.225-0.284
Matbench band-gap main case	Matbench v0.1 matbench_expt_gap	MAE=0.4513±0.0170 eV, RMSE=0.8049±0.0456 eV, R ² =0.6854±0.0445; greedy Recall@150=0.2626

S3.1 Regression workflows

For ESOL, the front end performed scaffold-aware five-fold cross-validation, selected RandomForest, and generated frozen predictions for 254 unlabeled test molecules; scoring against withheld public labels gave MAE = 0.603, RMSE = 0.741 and R² = 0.739. For Matbench steels, the front end processed official fold 0 and generated frozen predictions for 63 unlabeled test alloys; scoring gave MAE = 111.444 MPa, RMSE = 164.320 MPa and R² = 0.684. The Matbench steels result is reported as a single-fold front-end case.

S3.2 Extraction and retrieval workflows

Reaction-table extraction used the publisher JATS XML table as gold and evaluated 105 cells (21 rows). Exact precision, recall and F1 were each 0.981 and row accuracy was 0.905. The MassBank workflow ranked candidate molecular identities for five queries using public InChIKey labels, giving top-1 accuracy 0.800, top-3 and top-5 accuracy 1.000, and MRR 0.900. ChEMU Task 1 sample v3 contained 92 gold spans; exact precision/recall/F1 were 0.931/0.728/0.817 and relaxed F1 was 0.841.

S4. Retrospective BACE fixed-budget active discovery

The public library contained 1,513 molecules. High-activity targets were the top 5% by pIC₅₀ (n = 79). Each of 20 seeds began with 30 random observations and queried batches of 10 to a total budget of 150. The surrogate was ExtraTreesRegressor(n_estimators=200, max_depth=15). Model fitting and acquisition used queried observations, and unqueried labels served as the retrospective oracle for endpoint scoring.

Table S5. BACE fixed-budget endpoint metrics (mean ± SD across 20 seeds).

Policy	Recall@150	Hit rate@150	EF@150	Best pIC ₅₀ @150
random	0.109 ± 0.040	0.057 ± 0.021	1.098 ± 0.398	9.129 ± 0.491
greedy surrogate	0.333 ± 0.085	0.175 ± 0.045	3.358 ± 0.857	9.365 ± 0.583
ucb surrogate	0.363 ± 0.056	0.191 ± 0.030	3.664 ± 0.570	9.559 ± 0.632

Paired bootstrap intervals used 10,000 resamples of per-seed endpoint differences. Greedy-surrogate minus random Recall@150 had a 95% CI of 0.189–0.260; UCB minus random had a 95% CI of 0.225–0.284.

S5. Matbench experimental band-gap case

The archived front-end case used Matbench v0.1 matbench_expt_gap (n = 4,604) with 98 precomputed composition descriptors. HistGradientBoostingRegressor was evaluated by KFold(n_splits=5, shuffle=True,

random_state=18012019). The retrospective active-discovery component defined the top 5% experimental-gap materials as targets and used three seeds, 30 initial random observations, batches of 10 and a total budget of 150.

Table S6. Per-fold Matbench experimental band-gap regression metrics.

Fold	MAE (eV)	RMSE (eV)	R ²
0	0.4585	0.8223	0.6224
1	0.4261	0.7488	0.7547
2	0.4756	0.8819	0.6659
3	0.4397	0.7741	0.7105
4	0.4568	0.7974	0.6736

Across folds, MAE was 0.4513 ± 0.0170 eV, RMSE was 0.8049 ± 0.0456 eV, and R² was 0.6854 ± 0.0445 . At 150 queries, random and greedy-surrogate Recall@150 were 0.0332 and 0.2626, respectively. The paired recall improvement was 0.2294 with bootstrap 95% CI 0.2208–0.2381.

S6. BACE1 temporal external validation

S6.1 Data split and phase gate

ChEMBL target ChEMBL4822 records were filtered to standardized BACE1 IC50 measurements and aggregated by compound. Stage 1 used 5,296 compounds associated with documents published through 2015. Records from 2016–2017 were held as a temporal buffer. The external set contained 1,598 compounds associated with documents from 2018 onward after exact molecule overlap removal by InChIKey; 1,275 compounds were scaffold-novel. Stage 1 predictions and the prediction manifest were written before merging the external gold file.

Table S7. Stage-1 scaffold-grouped cross-validation metrics.

Model	Fold	MAE	RMSE	R ²
Ridge	0	0.6640	0.8524	0.4713
Ridge	1	0.6736	0.8675	0.4657
Ridge	2	0.6910	0.8961	0.4621
Ridge	mean	0.6762	0.8720	0.4664
Ridge	std	0.0137	0.0222	0.0046
RandomForest	0	0.5701	0.7415	0.5999
RandomForest	1	0.5642	0.7368	0.6146
RandomForest	2	0.6018	0.7910	0.5809
RandomForest	mean	0.5787	0.7564	0.5985
RandomForest	std	0.0202	0.0300	0.0169
ExtraTrees	0	0.6292	0.8362	0.4911
ExtraTrees	1	0.6201	0.8339	0.5063

Model	Fold	MAE	RMSE	R ²
ExtraTrees	2	0.6348	0.8373	0.5304
ExtraTrees	mean	0.6280	0.8358	0.5093
ExtraTrees	std	0.0074	0.0017	0.0198

RandomForest was selected by mean validation MAE and refit on all historical compounds using random seed 42. The model used Morgan FP radius=2, 1024bits + 8 RDKit descriptors and 100 trees with n_jobs = 1. Stage-1 runtime was 436.16 s.

S6.2 Independent metric recomputation

Table S8. Post-2018 external regression performance.

Model	MAE	RMSE	R ²	Pearson r
RandomForest	0.8215	1.0278	0.3687	0.6690
Ridge	0.9534	1.2235	0.1055	0.5938
Mean Baseline	1.1845	1.3712	-0.1234	-0.0000

Table S9. Prespecified top-decile recovery at a 10% selection budget.

Subset	n	k	Actives	Hits	Recall	EF
all external	1598	160	161	42	0.2609	2.6054
known scaffold	323	33	33	14	0.4242	4.1524
novel scaffold	1275	128	142	35	0.2465	2.4552

Independent local recomputation from external_predictions_with_gold.csv reproduced the selected-model values: MAE = 0.8214935, RMSE = 1.0278495 and R² = 0.3687357. The file contained 1,598 unique compound_id values, no missing values and a zero maximum absolute difference between Stage-1 selected predictions and the predictions carried into the Stage-2 merged file.

S6.3 Frozen-prediction integrity audit

The Stage-1 manifest records SHA-256 f2589c916135b0a934ac4ed23ec18b73b7cec99d2d94c686113fe325fece6e24. The HTTP artifact download used LF line endings and had raw SHA-256 f66bca3fce59791c60a2ae402fbdf3dfb3b40072850c036fd123afdf59648c4. Line-ending conversion from LF to CRLF yielded SHA-256 f2589c916135b0a934ac4ed23ec18b73b7cec99d2d94c686113fe325fece6e24, matching the manifest. The raw HTTP file and the manifest-tracked file therefore differ in line-ending format. Semantic prediction identity was verified by a one-to-one compound_id merge and zero maximum absolute prediction difference.

S7. Reproducibility, software and artifact provenance

S7.1 Explicitly enabled chemistry and materials Skills

The repository contained 147 discoverable SKILL.md files at package finalization. Table S10 reports the six chemistry/materials Skills explicitly marked as enabled in extensions_config.json. In this system, a Skill specifies instruction and tool-routing behavior; scientific foundations are therefore assigned to the underlying software or databases used for each scientific operation. The complete repository inventory is supplied as source_data/skills_inventory.csv.

Table S10. Enabled core domain Skills and their scientific foundations.

Skill	Configured role	Skill file under skills/public/	Scientific foundation
chemistry-studio-ketcher	Molecule and reaction canvas preparation; route-draft support	chemistry-studio-ketcher/SKILL.md	Ketcher [S9]; RDKit [S10]
chemistry-structure-resolution	Name/identifier resolution before downstream analysis	chemistry-structure-resolution/SKILL.md	PubChem [S14]; RDKit [S10]
rdkit-molecular-analysis	Descriptors, similarity, standardization, scaffolds, reaction QC and conformers	rdkit-molecular-analysis/SKILL.md	RDKit [S10]
materials-core	Formula, structure, XRD and precursor calculations	materials-core/SKILL.md	pymatgen [S11]
materials-evidence-project	Database-backed stability, band-gap and formation-energy evidence	materials-evidence-project/SKILL.md	Materials Project [S12]; OPTIMADE [S13]; PubChem [S14]
science-upload-autopilot	Routing of uploaded scientific files to deterministic tools with evidence-gap reporting	science-upload-autopilot/SKILL.md	This work; delegates to [S10-S14]

S7.2 Complete Skill catalogue

Table S11 lists every SKILL.md definition discovered under skills/public and skills/custom. The configuration label distinguishes the six explicitly enabled records from catalogue entries with no enablement record. Catalogue entries identify available instruction modules. Evaluation records specify experimental use, dependency availability, and validation status. References for the six chemistry/materials Skills are given in Table S10; other entries retain their declared license or repository provenance.

Table S11. Complete repository Skill catalogue at package finalization ($n = 147$).

Skill	Class	Configuration	Configured capability	Declared source or license
academic-paper	custom	catalogued	Manuscript drafting and revision workflow covering planning, outline development, literature review, citation checks, format conversion, and disclosure checks.	local custom skill

Skill	Class	Configuration	Configured capability	Declared source or license
academic-paper-reviewer	custom	catalogued	Structured manuscript review workflow with multiple review perspectives and field-specific checklists.	local custom skill
academic-pipeline	custom	catalogued	End-to-end manuscript workflow covering research organization, drafting, integrity checks, review, revision, and finalization.	local custom skill
nature-academic-search	custom	catalogued	Literature search and reference-management workflow covering citation verification, MeSH terms, citation files, and bibliography formats.	local custom skill
nature-citation	custom	catalogued	Citation-support workflow for manuscript passages, with source searching and segmentation for reference assignment.	local custom skill
nature-data	custom	catalogued	Data availability and FAIR metadata preparation workflow for manuscript supporting materials.	local custom skill
nature-figure	custom	catalogued	Manuscript figure preparation and audit workflow for Python/R-generated figures and multi-panel layouts.	local custom skill
nature-paper2ppt	custom	catalogued	Scientific presentation-generation workflow from papers, PDFs, abstracts, figure legends, and reading notes.	local custom skill
nature-polishing	custom	catalogued	Academic prose editing, restructuring, and translation workflow based on author-provided text and evidence.	local custom skill
nature-reader	custom	catalogued	Chinese-English paper-reading workflow for PDFs, DOIs, arXiv records, publisher HTML, and uploaded manuscripts.	local custom skill
nature-response	custom	catalogued	Point-by-point response-letter drafting and audit workflow for manuscript revisions.	local custom skill
nature-writing	custom	catalogued	Manuscript section drafting and restructuring workflow from author-provided claims, results, figures, and notes.	local custom skill
academic-paper-review	public	catalogued	Supports review, analyze, critique, or summarize academic papers, research articles, preprints, or scientific publications. Supports comprehensive...	repository skill definition

Skill	Class	Configuration	Configured capability	Declared source or license
adme-prediction	public	catalogued	ADME property prediction for absorption, distribution, metabolism and excretion endpoints, including Caco-2 permeability, PAMPA, HIA, Pgp inhibition, bioavailability and lipophilicity, using Morgan fingerprints with Random Forest/XGBoost backends.	repository skill definition
aeon	public	catalogued	Supports time series machine learning tasks including	BSD-3-Clause license
astropy	public	catalogued	Comprehensive Python library for astronomy and astrophysics. This skill	BSD-3-Clause license
bgpt-paper-search	public	catalogued	Search scientific papers and retrieve structured experimental data extracted	MIT
bootstrap	public	catalogued	Guided onboarding workflow for creating a personalized SOUL.md assistant profile.	repository skill definition
boron-nmr-predict	public	catalogued	Predict 11B (boron-11) NMR chemical shift for boron-containing molecules	repository skill definition
chart-visualization	public	catalogued	This skill should be used when the user wants to visualize data. It intelligently selects the most suitable chart type from 26 available options, extracts parameters based on...	repository skill definition
chem-data-extractor	public	catalogued	Extract structured chemical compound characterization data from chemistry\	repository skill definition
chemical-file-converter	public	catalogued	Chemical file-format conversion among .xyz, .gjf (Gaussian), .mol, .sdf, .pdb and .mol2 formats.	repository skill definition
chemistry-structure-resolution	public	explicitly enabled	Resolve chemical names, Chinese compound names, IUPAC names, common names, CAS-like identifiers, PubChem identifiers, SMILES, InChI, and ambiguous molecule descriptions into...	repository skill definition
chemistry-studio-ketcher	public	explicitly enabled	Natural-language Chemistry Studio workflow for drawing molecules, reactions, retrosynthesis sketches, and route drafts in Ketcher. Supports workflows to draw, load, sketch...	repository skill definition
cirq	public	catalogued	Google quantum computing framework. Supports workflows targeting Google Quantum	Apache-2.0 license
citation-management	public	catalogued	Comprehensive citation management for academic research. Search Google	MIT License
claude-to-deerflow	public	catalogued	Interact with DeerFlow AI agent platform via its HTTP API. Supports workflows in which the user wants to send messages or questions to DeerFlow for research/analysis, start a DeerFlow...	repository skill definition

Skill	Class	Configuration	Configured capability	Declared source or license
code-documentation	public	catalogued	Supports generate, create, or improve documentation for code, APIs, libraries, repositories, or software projects. Supports README generation, API...	repository skill definition
consciousness-council	public	catalogued	Run a multi-perspective Mind Council deliberation on any question, decision,	MIT license
consulting-analysis	public	catalogued	Supports generate, create, or write professional research reports including but not limited to market analysis, consumer insights, brand analysis...	repository skill definition
dask	public	catalogued	Distributed computing for larger-than-RAM pandas/NumPy workflows. Use	BSD-3-Clause license
data-analysis	public	catalogued	Supports workflows in which the user uploads Excel (.xlsx/.xls) or CSV files and wants to perform data analysis, generate statistics, create summaries, pivot tables, SQL queries, or any...	repository skill definition
database-lookup	public	catalogued	Search 78 public scientific, biomedical, materials science, and economic	repository skill definition
deep-research	public	catalogued	Web-research workflow for information retrieval, source extraction, and multi-step synthesis.	repository skill definition
docx	public	catalogued	Supports create, read, edit, or manipulate	Proprietary. LICENSE.txt has complete terms
exploratory-data-analysis	public	catalogued	Perform comprehensive exploratory data analysis on scientific data files	MIT license
find-skills	public	catalogued	Helps users discover and install agent skills when they ask questions like "how do I do X", "find a skill for X", "is there a skill that can...", or express interest in extending...	repository skill definition
fluidsim	public	catalogued	Framework for computational fluid dynamics simulations using Python.	CeCILL FREE SOFTWARE LICENSE AGREEMENT
frontend-design	public	catalogued	Frontend interface development workflow for web components, pages, artifacts, posters and related visual layouts.	Complete terms in LICENSE.txt
generate-image	public	catalogued	Generate or edit images using AI models (FLUX, Nano Banana 2). Use for	MIT license
geomaster	public	catalogued	Comprehensive geospatial science skill covering remote sensing, GIS,	MIT License

Skill	Class	Configuration	Configured capability	Declared source or license
geometry-optimizer	public	catalogued	Semi-empirical molecular geometry optimization with xTB; supports SMILES-to-3D conversion and XYZ input, and returns optimized coordinates, energy and convergence status.	repository skill definition
geopandas	public	catalogued	Python library for working with geospatial vector data including shapefiles,	BSD-3-Clause license
get-available-resources	public	catalogued	This skill should be used at the start of any computationally intensive	MIT license
github-deep-research	public	catalogued	Conduct multi-round deep research on any GitHub Repo. Supports comprehensive analysis, timeline reconstruction, competitive analysis, or in-depth investigation of...	repository skill definition
gjf-to-xyz	public	catalogued	Convert Gaussian gjf input files to XYZ format.	repository skill definition
hypogenic	public	catalogued	Automated LLM-driven hypothesis generation and testing on tabular datasets.	MIT license
hypothesis-generation	public	catalogued	Structured hypothesis formulation from observations. Supports workflows where you have	MIT license
image-generation	public	catalogued	Supports generate, create, imagine, or visualize images including characters, scenes, products, or any visual content. Supports structured prompts...	repository skill definition
infographics	public	catalogued	Create professional infographics using Nano Banana Pro AI with smart	repository skill definition
ir-spectra-simulation	public	catalogued	Compute and visualize IR Spectra from input SMILES or XYZ file with MLatom.	repository skill definition
iso-13485-certification	public	catalogued	Comprehensive toolkit for preparing ISO 13485 certification documentation	MIT license
iupac-to-smiles	public	catalogued	IUPAC-name-to-SMILES conversion using the OPSIN API, with support for polymer-oriented parsing.	repository skill definition
labarchive-integration	public	catalogued	Electronic lab notebook API integration. Access notebooks, manage entries/attachments,	Unknown
latex-posters	public	catalogued	Create professional research posters in LaTeX using beamerposter, tikzposter,	repository skill definition
literature-parsing	public	catalogued	Convert PDF literature to Markdown using MinerU API. GVIM/DeerFlow uses API mode by default, so the local mineru package is not required.	repository skill definition
literature-review	public	catalogued	Conduct comprehensive, systematic literature reviews using multiple academic	MIT license

Skill	Class	Configuration	Configured capability	Declared source or license
markdown-mermaid-writing	public	catalogued	Comprehensive markdown and Mermaid diagram writing skill. Supports workflows where creating	Apache-2.0
market-research-reports	public	catalogued	Generate comprehensive market research reports (50+ pages) in the style	MIT license
markitdown	public	catalogued	Convert files and office documents to Markdown. Supports PDF, DOCX, PPTX,	MIT license
matchms	public	catalogued	Spectral similarity and compound identification for metabolomics. Use	Apache-2.0 license
materials-core	public	explicitly enabled	Package-backed materials science workflows for formulas, structures, XRD, precursor stoichiometry, units, spectra, local environments, and composition descriptors. Supports workflows where the...	repository skill definition
materials-evidence-project	public	explicitly enabled	Evidence-first materials database workflow for Materials Project, OPTIMADE, PubChem reference records, X-ray references, scattering references, stability, band gaps, formation...	repository skill definition
matlab	public	catalogued	MATLAB and GNU Octave numerical computing for matrix operations, data	For MATLAB (https://www.mathworks.com/pricing-licensing.html) and for Octave
matplotlib	public	catalogued	Low-level plotting library for full customization. Supports workflows where you need	https://github.com/matplotlib/matplotlib/tree/main/LICENSE
md-vib-spectra-simulation	public	catalogued	Compute vibrational spectra from MD trajectory — power spectrum (velocity	repository skill definition
modal	public	catalogued	Cloud computing platform for running Python on GPUs and serverless infrastructure.	Apache-2.0
mol-2d-viewer	public	catalogued	Conversion of SMILES strings or chemical names into 2D molecular structure drawings, including common molecules and polymers.	repository skill definition
mol-3d-viewer	public	catalogued	Conversion of SMILES strings or chemical names into 3D molecular structures; supports SDF output, static 3D images and interactive HTML views.	repository skill definition
mol-image-to-smiles	public	catalogued	Molecular-structure-image recognition and conversion to SMILES using DECIMER and MolNextR models.	repository skill definition

Skill	Class	Configuration	Configured capability	Declared source or license
mol-paper-renderer	public	catalogued	Publication-oriented molecular rendering with xyzrenderer, supporting SVG, PNG, PDF and GIF outputs for transition states, noncovalent interactions, molecular orbitals and crystal structures.	repository skill definition
molecular-dynamics	public	catalogued	Run and analyze molecular dynamics simulations with OpenMM and MDAnalysis.	MIT
molecular-properties-predictor	public	catalogued	Small-molecule physicochemical property prediction for boiling point, refractive index, density, viscosity, surface tension and related endpoints, using the bamboo_mixer backend.	repository skill definition
ms-spectra-simulation	public	catalogued	Predict and visualize MS/MS spectra from a single SMILES using the fioRa	repository skill definition
networkx	public	catalogued	Comprehensive toolkit for creating, analyzing, and visualizing complex	3-clause BSD license
newsletter-generation	public	catalogued	Supports generate, create, write, or draft a newsletter, email digest, weekly roundup, industry briefing, or curated content summary. Supports...	repository skill definition
nmr-prediction	public	catalogued	Predict liquid-phase ^1H and ^{13}C NMR chemical shifts from a SMILES string	repository skill definition
open-notebook	public	catalogued	Self-hosted, open-source alternative to Google NotebookLM for AI-powered	MIT
opentrons-integration	public	catalogued	Official Opentrons Protocol API for OT-2 and Flex robots. Supports workflows where writing	Unknown
optimize-for-gpu	public	catalogued	GPU-accelerate Python code using CuPy, Numba CUDA, Warp, cuDF, cuML,	repository skill definition
paper-lookup	public	catalogued	Search 10 academic paper databases via their REST APIs to find research	repository skill definition
paperzilla	public	catalogued	Chat with your agent about projects, recommendations, and canonical papers	MIT
parallel-web	public	catalogued	Search the web, extract URL content, and run deep research using the	MIT license
pdf	public	catalogued	Supports do anything with PDF files.	Proprietary. LICENSE.txt has complete terms
pdf-dft-extractor	public	catalogued	Extract DFT coordinate blocks from PDF supplementary information via MinerU API and generate Gaussian .gif files.	repository skill definition
peer-review	public	catalogued	Structured manuscript/grant review with checklist-based evaluation. Use	MIT license

Skill	Class	Configuration	Configured capability	Declared source or license
pennylane	public	catalogued	Hardware-agnostic quantum ML framework with automatic differentiation.	Apache-2.0 license
perplexity-search	public	catalogued	Perform AI-powered web searches with real-time information using Perplexity	MIT license
pka-predictor	public	catalogued	Small-molecule pKa prediction using a custom heuristic backend and a Uni-pKa single-file-weight backend.	repository skill definition
podcast-generation	public	catalogued	Supports generate, create, or produce podcasts from text content. Converts written content into a two-host conversational podcast audio format...	repository skill definition
polars	public	catalogued	Fast in-memory DataFrame library for datasets that fit in RAM. Use when	https://github.com/pola-rs/polars/blob/main/LICENSE
ppt-generation	public	catalogued	Supports generate, create, or make presentations (PPT/PPTX). Creates visually rich slides by generating images for each slide and composing them...	repository skill definition
pptx	public	catalogued	Use this skill any time a .pptx file is involved in any way — as input,	Proprietary. LICENSE.txt has complete terms
pptx-posters	public	catalogued	Create research posters using HTML/CSS that can be exported to PDF or	MIT license
protocolsio-integration	public	catalogued	Integration with protocols.io API for managing scientific protocols.	Unknown
pufferlib	public	catalogued	High-performance reinforcement learning framework optimized for speed	MIT license
pylabrobot	public	catalogued	Vendor-agnostic lab automation framework. Supports workflows where controlling multiple	MIT license
pymatgen	public	catalogued	Materials science toolkit. Crystal structures (CIF, POSCAR), phase diagrams,	MIT license
pymc	public	catalogued	Bayesian modeling with PyMC. Build hierarchical models, MCMC (NUTS),	Apache License, Version 2.0
pymoo	public	catalogued	Multi-objective optimization framework. NSGA-II, NSGA-III, MOEA/D, Pareto	Apache-2.0 license
pyopenms	public	catalogued	Complete mass spectrometry analysis platform. Supports proteomics workflows	3 clause BSD license
pytorch-lightning	public	catalogued	Deep learning framework (PyTorch Lightning). Organize PyTorch code into	Apache-2.0 license
pyzotero	public	catalogued	Interact with Zotero reference management libraries using the pyzotero	MIT License

Skill	Class	Configuration	Configured capability	Declared source or license
qiskit	public	catalogued	IBM quantum computing framework. Supports workflows targeting IBM Quantum hardware,	Apache-2.0 license
qutip	public	catalogued	Quantum physics simulation library for open quantum systems. Use when	BSD-3-Clause license
raman-spectra-simulation	public	catalogued	Compute and visualize Raman Spectra from input SMILES or XYZ file with	repository skill definition
rdkit	public	catalogued	Cheminformatics toolkit for fine-grained molecular control. SMILES/SDF	BSD-3-Clause license
rdkit-molecular-analysis	public	explicitly enabled	RDKit-backed molecular analysis for descriptors, similarity, standardization, scaffolds, fragments, substructure checks, reaction QC, and 3D conformer generation. Use for...	repository skill definition
reaction-data-extraction	public	catalogued	Extract chemistry reaction tables and reaction conditions from PDF literature using MinerU API plus rule-based parsing.	repository skill definition
research-grants	public	catalogued	Write competitive research proposals for NSF, NIH, DOE, DARPA, and Taiwan	MIT license
research-lookup	public	catalogued	Look up current research information using the Parallel Chat API (primary)	MIT license
scholar-evaluation	public	catalogued	Systematically evaluate scholarly work using the ScholarEval framework,	MIT license
science-upload-autopilot	public	explicitly enabled	Auto-route uploaded or pasted chemistry/materials data such as CIF, POSCAR, XYZ, CSV formula tables, SMILES lists, XRD peak tables, and XY spectra through GVIM science tools...	repository skill definition
scientific-brainstorming	public	catalogued	Creative research ideation and exploration. Supports open-ended brainstorming	MIT license
scientific-critical-thinking	public	catalogued	Evaluate scientific claims and evidence quality. Supports assessing experimental	MIT license
scientific-schematics	public	catalogued	Create publication-quality scientific diagrams using Nano Banana 2 AI	MIT license
scientific-slides	public	catalogued	Build slide decks and presentations for research talks. Use this for	MIT license
scientific-visualization	public	catalogued	Meta-skill for publication-ready figures. Supports workflows where creating journal submission	MIT license
scientific-writing	public	catalogued	Core skill for the deep research and writing tool. Write scientific manuscripts	MIT license
scikit-learn	public	catalogued	Machine learning in Python with scikit-learn. Supports workflows where working with supervised	BSD-3-Clause license
seaborn	public	catalogued	Statistical visualization with pandas integration. Supports quick exploration	BSD-3-Clause license

Skill	Class	Configuration	Configured capability	Declared source or license
shap	public	catalogued	Model interpretability and explainability using SHAP (SHapley Additive)	MIT license
simpy	public	catalogued	Process-based discrete-event simulation framework in Python. Use this	MIT license
skill-creator	public	catalogued	Create new skills, modify and improve existing skills, and measure skill performance. Supports workflows where users want to create a skill from scratch, edit, or optimize an existing skill, run...	repository skill definition
smiles-to-iupac	public	catalogued	SMILES-to-IUPAC-name conversion using PubChem, NCI/CADD, STOUT and RDKit-InChI routes.	repository skill definition
stable-baselines3	public	catalogued	Production-ready reinforcement learning algorithms (PPO, SAC, DQN, TD3,	MIT license
statistical-analysis	public	catalogued	Guided statistical analysis with test selection and reporting. Use when	MIT license
statsmodels	public	catalogued	Statistical models library for Python. Supports workflows where you need specific model	BSD-3-Clause license
surface-tension-predictor	public	catalogued	SMILES-based reference prediction for surfactant-related surface tension tasks, with baseline heuristic and public_joblib backends.	repository skill definition
surprise-me	public	catalogued	Exploratory workflow that combines enabled Skills to demonstrate available system capabilities.	repository skill definition
sympy	public	catalogued	Supports workflows in which working with symbolic mathematics in Python. This	https://github.com/sympy/sympy/blob/master/LICENSE
systematic-literature-review	public	catalogued	Supports workflows in which the user wants a systematic literature review, survey, or synthesis across multiple academic papers on a topic. Also covers annotated bibliographies and...	repository skill definition
timesfm-forecasting	public	catalogued	Zero-shot time series forecasting with Google's TimesFM foundation model.	Apache-2.0 license
torch-geometric	public	catalogued	Guide for building Graph Neural Networks with PyTorch Geometric (PyG).	repository skill definition
transformers	public	catalogued	This skill should be used when working with pre-trained transformer models	Apache-2.0 license
umap-learn	public	catalogued	UMAP dimensionality reduction. Fast nonlinear manifold learning for 2D/3D	BSD-3-Clause license
usfiscaldata	public	catalogued	Query the U.S. Treasury Fiscal Data API for federal financial data including	MIT

Skill	Class	Configuration	Configured capability	Declared source or license
uv-vis-spectrum-simulation	public	catalogued	UV-Vis spectrum from SMILES via UV-adVISor (https://spectra.collaborationspharma.com/).	repository skill definition
vaex	public	catalogued	Use this skill for processing and analyzing large tabular datasets (billions)	MIT license
venue-templates	public	catalogued	Access comprehensive LaTeX templates, formatting requirements, and submission	MIT license
vercel-deploy	public	catalogued	Deploy applications and websites to Vercel. Supports workflows in which the user requests deployment actions such as "Deploy my app", "Deploy this to production", "Create a preview..."	repository skill definition
video-generation	public	catalogued	Supports generate, create, or imagine videos. Supports structured prompts and reference image for guided generation.	repository skill definition
web-design-guidelines	public	catalogued	Review UI code for Web Interface Guidelines compliance. Supports requests to "review my UI", "check accessibility", "audit design", "review UX", or "check my site against best..."	repository skill definition
what-if-oracle	public	catalogued	Run structured What-If scenario analysis with multi-branch possibility	MIT license
xlsx	public	catalogued	Use this skill any time a spreadsheet file is the primary input or output.	Proprietary. LICENSE.txt has complete terms
xrd-spectra-simulation	public	catalogued	XRD spectrum from CIF structure file using pymatgen (Cu K α).	repository skill definition
zarr-python	public	catalogued	Chunked N-D arrays for cloud storage. Compressed arrays, parallel I/O,	MIT license

S7.3 Software versions and artifact provenance

Table S12. Software recorded by the BACE1 temporal Stage-1 manifest.

Component	Version
python	3.12.9 (main, Feb 12 2025, 14:52:31) [MSC v.1942 64 bit (AMD64)]
numpy	2.4.6
pandas	2.3.3
scikit-learn	1.8.0
rdkit	2026.03.2

Table S13. Key local artifacts and SHA-256 identifiers.

Artifact	SHA-256
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Artifact	SHA-256
research-demos/results/fast_ml_showcase/frontend_esol_scaffold_aware.json	093daf7b8f53f2a8aad7f5e9bf562a16bfd43eb4c7c250e88cfcf80a0da9edf5
research-demos/results/fast_ml_showcase/frontend_steels_single_run_recheck.json	2e6df9eae07be3dac1fbbf6b69183f5a1b6b3e38db67639942a747ecaa528950
research-demos/results/matbench_steels_5fold/independent_evaluation.json	d51de3b172bee43067aced669fa533ee1fbf8c9896390b64757429d3eb695c6e
research-demos/results/mineru_reaction_table/frontend_9dc67557_evaluation.json	b548fcfc82598ca423fb1b94e62cf3386246b3b8d191b49757d7659b3ee58583
research-demos/results/msms_retrieval/frontend_abe44a97_evaluation.json	b313f8461f29818d4a4ebf623e7b0626e4d3f2829ce34acd77db2ab613754a1c
research-demos/results/chemu/frontend_dbb54347_evaluation.json	1d77462d9fccd1b085305dc8963cf5fcfb3ce9ff1de6a12fd708f16392f8fb5
research-demos/results/frontend_bace_fff9cae7/metrics.json	1672a1884ccea0c053cbd705879ee7334d2543364687b8bd76f354a082659bcd
research-demos/results/frontend_fast_features_a6600dc1/metrics.json	47fdcd6fdd5b9326fd49428f4123074afaf49df45aa27ffa3591cb64589ddff92
research-demos/results/frontend_bace_temporal_9ae4e85f/metrics.json	01baf6f4c349a4cc4f3022d627101a7eba259330e9fdd2e06b57803a6fce19c
research-demos/results/frontend_bace_temporal_9ae4e85f/prediction_manifest.json	bf1f4355d64df56e25d9843d07d4b38f33a4df3ab7c2c98767e067f921effccb

The temporal-validation front-end thread identifier was 9ae4e85f-542b-4f94-9294-81189cf220be. Archived front-end identifiers also include 9dc67557-7d82-476d-80e0-381b7af1e76f for reaction-table extraction, abe44a97-e5a9-4ba3-bbf3-6206ab9d7c12 for MS/MS retrieval, dbb54347-66bc-4244-8177-bdf441383454 for ChEMU, fff9cae7-3467-496d-af05-0585c99fd993 for BACE active discovery, and a6600dc1-8a72-46d1-97c0-d6ce0020c7f6 for the Matbench band-gap case.

S8. Metric definitions

- $\text{MAE} = n^{-1} \sum_i |y_i - \hat{y}_i|$.
- $\text{RMSE} = [n^{-1} \sum_i (y_i - \hat{y}_i)^2]^{1/2}$.
- $R^2 = 1 - \sum_i (y_i - \hat{y}_i)^2 / \sum_i (y_i - \bar{y})^2$.
- Precision = TP/(TP + FP), recall = TP/(TP + FN), and F1 is their harmonic mean.
- Top-k accuracy is the fraction of queries for which the gold identity appears among the first k ranked candidates; MRR is the mean reciprocal rank of the gold candidate.
- Recall@k is the fraction of prespecified target compounds recovered within k queries or selections.
- EF@k = (target fraction among selected compounds)/(target fraction in the full evaluation library).

S9. Additional methodological notes

- Retrospective discovery and temporal validation were evaluated with recall, enrichment and external prediction metrics based on public labels.

- Task-native extraction and retrieval cases were evaluated by fixed artifact-level endpoints, including F1, top-k accuracy and MRR.
- The 400-task aggregate combines benchmarks with different native scoring rules and is reported as a descriptive summary.
- The BACE1 temporal model retained non-random enrichment for scaffold-novel compounds, while external prediction errors increased and some highly active novel chemotypes were underpredicted.
- Component-level attribution can be examined through paired agent-baseline comparisons and component-ablation studies.

S10. Supporting references

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